

COMP3052 Computer Security

Session 02: Motivating Example

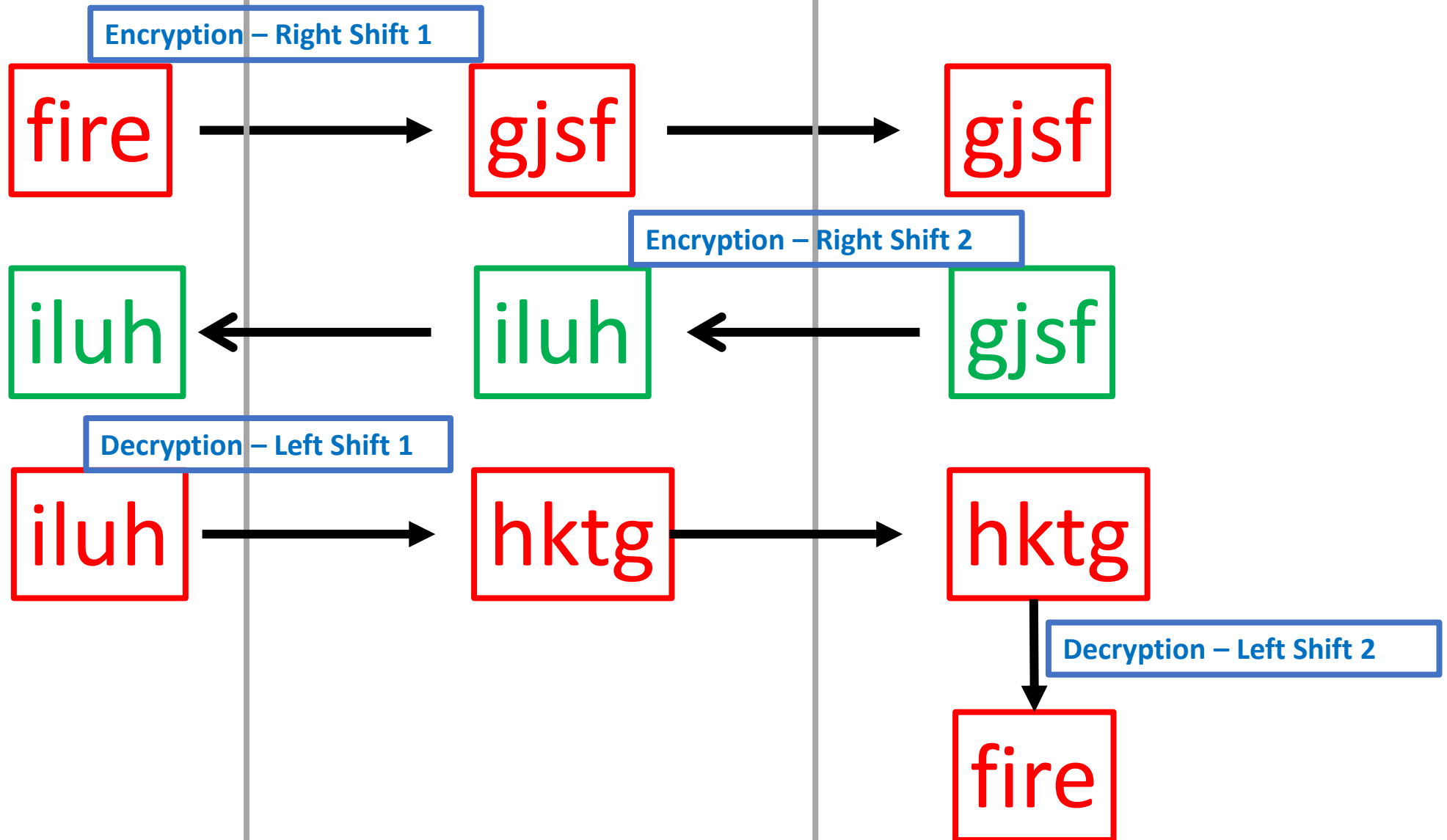
Videoclip: Key Exchange

<https://www.youtube.com/watch?v=U62S8SchxX4>

Videoclip: How Encryption Keys Work - with Chris Bishop

<https://www.youtube.com/watch?v=EDTx3meleT0>

Key Exchange



Generate Hash Values from Passwords using Hashing Algorithms (md5 & sha512)

```
drwxr-xr-x 2 sec sec 4096 Nov 17 2020 lab6
```

```
drwxr-xr-x 2 sec sec 4096 Nov 17 2020 lab7
```

```
sec@kali:~$
```

```
sec@kali:~$ cd lab4
```

Change directory to lab4

```
sec@kali:~/lab4$
```

```
sec@kali:~/lab4$ ls -l
```

```
total 32
```

```
-rwxr-xr-x 1 sec sec 371 Nov 17 2020 checkpasswd
```

```
drwxr-xr-x 2 sec sec 4096 Nov 17 2020 dicts
```

```
drwxr-xr-x 2 sec sec 4096 Nov 17 2020 hashes
```

```
drwxr-xr-x 3 sec sec 4096 Nov 17 2020 pack
```

```
-rwxr-xr-x 1 sec sec 118 Nov 17 2020 reset
```

```
drwxr-xr-x 3 sec sec 4096 Nov 17 2020 rules
```

```
-rwxr-xr-x 1 sec sec 431 Nov 17 2020 status
```

```
-rwxr-xr-x 1 sec sec 108 Nov 17 2020 submitpasswd
```

```
sec@kali:~/lab4$
```

```
sec@kali:~/lab4$ echo -n "password" | openssl md5
```

Hash value of 32 characters long.

```
(stdin)= 6d1ae5cf1c748c9125a422b9cf53c9d0
```

```
sec@kali:~/lab4$
```

```
sec@kali:~/lab4$ echo -n "password" | openssl sha512
```

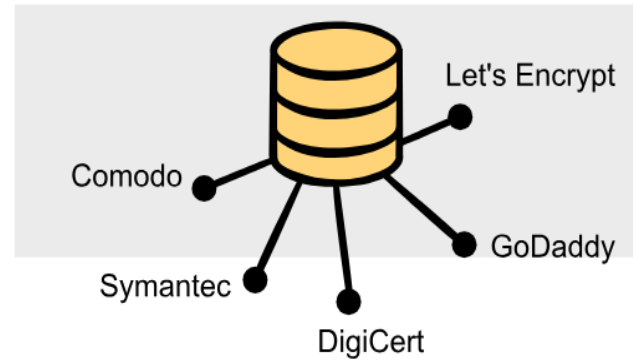
Encrypt the string "password" with the hashing algorithm (sha512).

```
(stdin)= 06195409b5f92bc9f64a2cfedeec634cbc3f8e9c4929f402d2eee05723606119e967c4e28a3822bfe47f94c683f3d02afb7a794daf284b1b1866faf829f33741
```

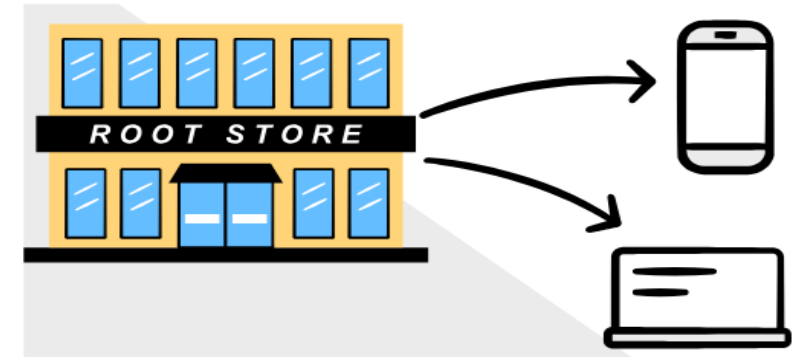
```
sec@kali:~/lab4$
```

```
sec@kali:~/lab4$
```

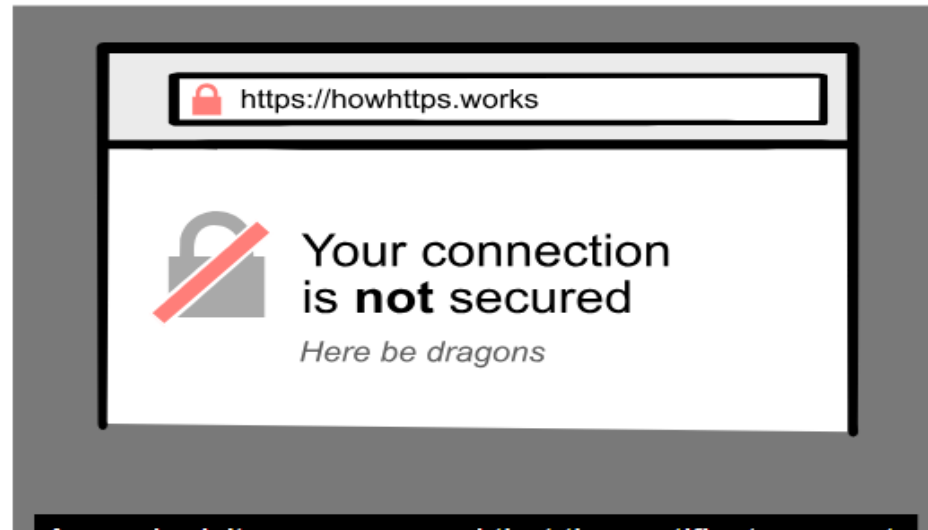
Hash value of 128 characters long.



A root store is basically a database of trusted CAs.

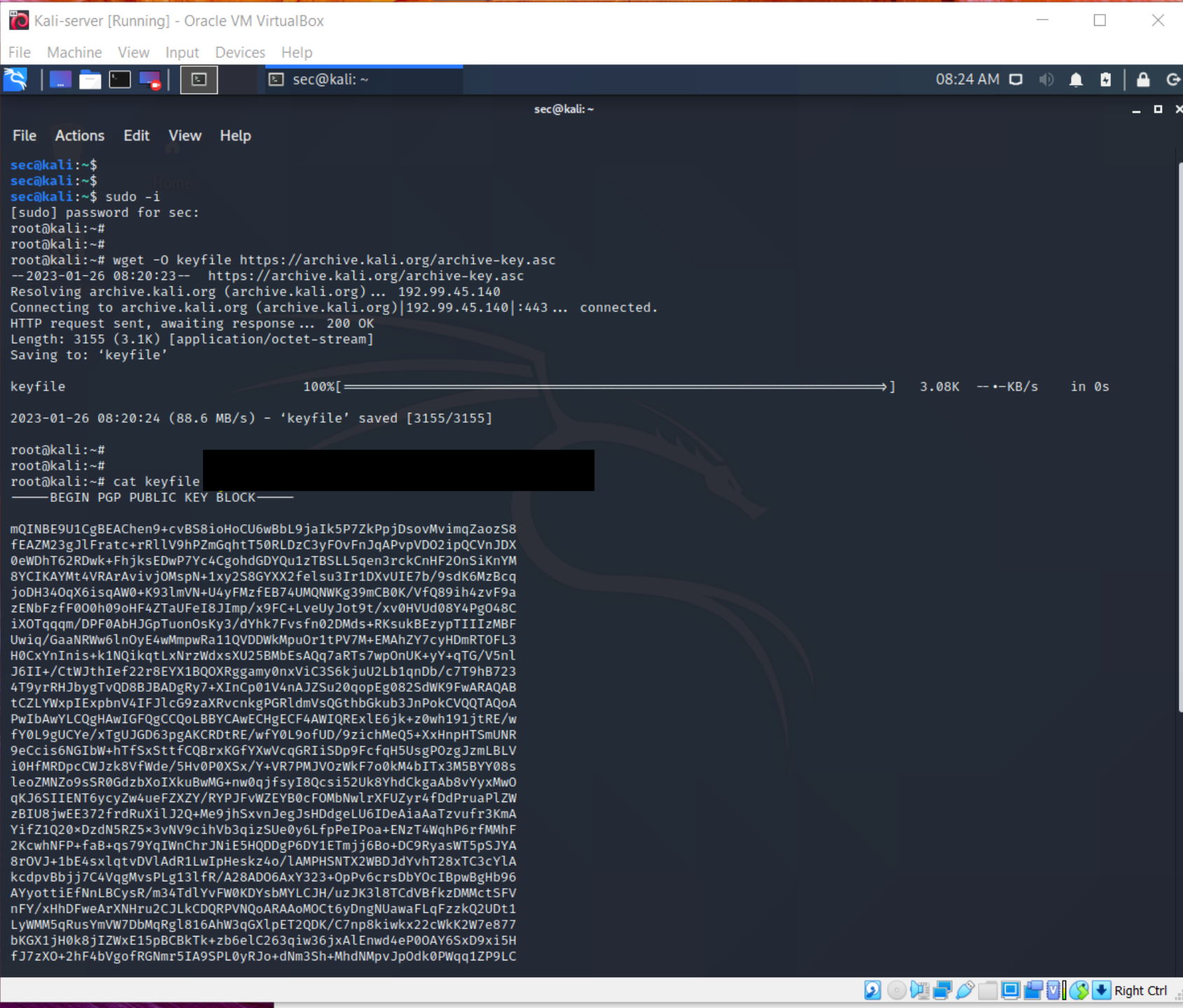


Apple, Windows, and Mozilla run their own root stores that they pre-install in your computer or device.



Reference: Certificate Authorities

<https://howhttps.works/certificate-authorities/>



1. Can someone spy on your data if your connection is not secured?
(A) Yes (B) No

2. Why do we need HTTPS?
(A) For privacy and identification (B) For faster websites
(C) For privacy, integrity, and identification (D) For identification only

3. In the context of HTTPS, what does integrity mean?
(A) That my browser has ethics
(B) That communication is not being tampered with
(C) That the website I am visiting is honest
(D) That the internet is strong and durable

Reference: The differences between HTTPS, SSL, and TLS

<https://howhttps.works/https-ssl-tls-differences/>

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<https://howhttps.works/https-ssl-tls-differences/>

1. In _____ cryptography, the same key is used by the sender (for encryption) and the receiver (for decryption).
(A) Symmetric-key (B) Asymmetric-key
(C) Public-key (D) None of the above

2. In _____ cryptography, everyone has access to everyone's public key.
(A) Symmetric-key (B) Asymmetric-key
(C) Both (A) and (B) (D) None of the above

Reference: Chapter 28 Quiz

https://highered.mheducation.com/sites/0072967722/student_view0/chapter_28_quiz.html

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1. What is hashing in the context of security?

Answer: A hash function is any function that can be used to map data of arbitrary size to fixed-size values. The values returned by a hash function are called hash values, hash codes, digests, or simply hashes.

2. Why are hash functions used to store passwords in DB?

Answer: Since hash is a one-way compression function, instead of storing the password itself, its hash value is stored. The user enters their password at the time of login. Then the hash value of the entered password will be compared with the stored hash value in DB and if both matched then the user will be logged in.

Reference: 20 Web Security Questions

<https://circuit.bcit.ca/repository/islandora/object/repository%3A1362/datastream/PDF/view>

1. A _____ serves as the trusted third-party agency that is responsible for issuing the digital certificates.

Answer: Certificate Authority (CA)

Explanation: A CA is an entity that is responsible for issuing digital certificates. These certificates are used to verify the authenticity and integrity of digital communications and transactions. The CA acts as a trusted third-party agency, ensuring that the certificates are issued to the correct entities and that they can be trusted by relying parties. The CA uses cryptographic algorithms to generate and sign these certificates, providing a secure and reliable mechanism for establishing trust in the digital world.

Reference: How Much Do You Know About Digital Certificate? Chapter 12 Quiz

https://www.proprofs.com/quiz-school/story.php?title=chap-12_4

Reference: Digital Certificates Explained - How digital certificates bind owners to their public key

<https://www.youtube.com/watch?v=5rT6fZUwhG8>